Solar Fe I appendix

Sun · Fe I · downloadable appendix

Document	Solar Fe I appendix
Subject	Solar photospheric abundance appendix · Fe I
Author	Ryan Schmitt
Date	2026-11-06
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.575
Source	science commit 28f3e442ab20
Web	https://exoplanetcodex.org/

Fe.json v1.575 · 187 Fe I products · 242 total

Solar Fe I anchor: 7.477 ± 0.049

HARPS · Reference Grade · Amarsi 3D-NLTE · n = 21. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

Highlighted band products

near-UV

7.452 ± 0.065

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · n = 51

Uncertainty budget closed for six of eight rows (RYA-1226); two Fe I Kitt Peak/molecfit rows keep a legacy bar pending their microturbulence bracket. On the migrated rows most of the quoted systematic is a conservative profile-width term carried from a measurement in another band.

VIS

7.504 ± 0.046

Kitt Peak Molecfit · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

VIS

7.477 ± 0.049

HARPS · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

red-optical

7.423 ± 0.048

Kitt Peak · Synth · 1D-LTE · Reference Grade · n = 68

NIR

7.481 ± 0.048

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · n = 1

H

7.649 ± 0.053

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · n = 8

Line grades

Grade	What it means	This element
Reference Grade	An externally published reference line pool, using that reference's line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.	122 products — asplund; reference
Codex Grade	The vetted Codex pool of primary-laboratory oscillator strengths, restricted to lines at or below the adopted feature-depth gate.	56 products — our-graded
Deep Grade	The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.	9 products — our-deep-graded

Error-bar forest

How to read this forest. Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool's provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Yellow rows are Deep Grade: strong lines that saturate, kept but less certain than the other grades. Unmeasured model rows remain N/A.

Model labels

1D-LTE uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

3D-NLTE (Amarsi) includes three-dimensional atmospheric structure and non-LTE line formation; these products apply the published Amarsi correction grid to a 1D baseline.

Mean-3D / Frankenstein combines a spatially and temporally averaged 3D atmosphere with the Gerber synthesis route. Averaging removes the full 3D fluctuations and velocity field; this experimental hybrid is not a full 3D calculation or an adopted engine.

Synthesis fits the spectral line profile. **EW** infers abundance from a line's integrated absorption (equivalent width). See Methodology for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

bar = σ as Asplund 2021 quote it (scatter, 3D, NLTE) · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders

near-UV — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.27–7.78

Synth · 1D-LTE
Deep Grade · n=55



7.554
±0.075

near-UV — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.27–7.78

Synth · 1D-LTE Reference Grade · n=48		7.515 ±0.072
Synth · 1D-LTE · Gerber Reference Grade · n=51		7.494 ±0.067
Synth · 1D-NLTE · Bergemann Reference Grade · n=48		7.523 ±0.072
Synth · 1D-NLTE · Gerber Reference Grade · n=51		7.559 ±0.067
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=52		7.588 ±0.070
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=53		7.678 ±0.069

near-UV — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.27–7.78

Synth · 1D-LTE Reference Grade · n=46		7.380 ±0.075
Synth · 1D-LTE · Gerber Reference Grade · n=50		7.398 ±0.065
Synth · 1D-NLTE · Bergemann Reference Grade · n=46		7.389 ±0.075
Synth · 1D-NLTE · Gerber Reference Grade · n=51		7.452 ±0.065
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=51		7.500 ±0.067
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=51		7.543 ±0.062

near-UV axis: 7.27 · 7.40 · 7.53 · 7.65 · 7.78 A(Fe) dex

VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Reference Grade · n=22		7.443 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=22		7.450 ±0.048

VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-NLTE · Bergemann Reference Grade · n=18		7.462 ±0.049
Synth · 1D-NLTE · Gerber Reference Grade · n=22		7.484 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.531 ±0.048
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.546 ±0.049
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.471 ±0.046

VIS — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Deep Grade · n=106		7.329 ±0.048
Synth · 1D-NLTE · Bergemann Deep Grade · n=101		7.340 ±0.048

VIS — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.66

EW · 1D-LTE Codex Grade · n=13 · 1 alternate		7.445 ±0.104
Synth · 1D-LTE Codex Grade · n=65		7.410 ±0.050
Synth · 1D-LTE · Gerber Codex Grade · n=67		7.439 ±0.067
EW · 1D-NLTE · Bergemann Codex Grade · n=12		7.455 ±0.107
Synth · 1D-NLTE · Bergemann Codex Grade · n=59		7.411 ±0.050
Synth · 1D-NLTE · Gerber Codex Grade · n=67		7.488 ±0.066
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.541 ±0.067
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.562 ±0.066

VIS — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.505 ±0.047

VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Reference Grade · n=166		7.367 ±0.045
Synth · 1D-LTE · Gerber Reference Grade · n=167		7.373 ±0.044
Synth · 1D-NLTE · Bergemann Reference Grade · n=156		7.371 ±0.045
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.430 ±0.044
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=101		7.485 ±0.045

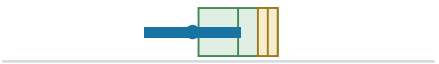
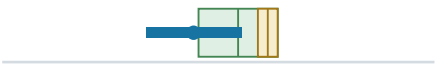
VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Reference Grade · n=22		7.455 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=22		7.462 ±0.048
Synth · 1D-NLTE · Bergemann Reference Grade · n=18		7.469 ±0.049
Synth · 1D-NLTE · Gerber Reference Grade · n=22		7.495 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.542 ±0.049
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.562 ±0.049
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.504 ±0.046

VIS — Kitt Peak — 1984 Molecfit corrected Codex deep pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Deep Grade · n=106		7.347 ±0.048
Synth · 1D-NLTE · Bergemann Deep Grade · n=102		7.357 ±0.048

VIS — Kitt Peak — 1984 Molecfit corrected Codex graded pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

EW · 1D-LTE Codex Grade · n=13		7.463 ±0.111
Synth · 1D-LTE Codex Grade · n=65		7.414 ±0.049
Synth · 1D-LTE · Gerber Codex Grade · n=65		7.415 ±0.048
EW · 1D-NLTE · Bergemann Codex Grade · n=12		7.464 ±0.112
Synth · 1D-NLTE · Bergemann Codex Grade · n=59		7.420 ±0.049
Synth · 1D-NLTE · Gerber Codex Grade · n=65		7.467 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=65		7.511 ±0.048
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=65		7.548 ±0.048
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.047

VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Reference Grade · n=167		7.368 ±0.044
Synth · 1D-LTE · Gerber Reference Grade · n=166		7.378 ±0.044
Synth · 1D-NLTE · Bergemann Reference Grade · n=157		7.377 ±0.045
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.432 ±0.044

VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=101		7.492 ±0.045

VIS — HARPS Reference Grade · Asplund replication solar_harps_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Reference Grade · n=27		7.398 ±0.047
Synth · 1D-LTE · Gerber Reference Grade · n=27		7.400 ±0.047
Synth · 1D-NLTE · Bergemann Reference Grade · n=20		7.417 ±0.049
Synth · 1D-NLTE · Gerber Reference Grade · n=27		7.452 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=27		7.480 ±0.048
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=27		7.515 ±0.048
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.477 ±0.049

VIS — HARPS Codex deep pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Deep Grade · n=106		7.419 ±0.051
Synth · 1D-NLTE · Bergemann Deep Grade · n=102		7.429 ±0.051

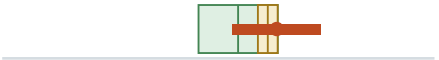
VIS — HARPS Codex deep pool · local continuum solar_harps_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Deep Grade · n=109		7.339 ±0.087
Synth · 1D-NLTE · Bergemann Deep Grade · n=105		7.350 ±0.088

VIS — HARPS Codex graded pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

EW · 1D-LTE Codex Grade · n=6 · 1 alternate		7.498 ±0.120

VIS — HARPS Codex graded pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

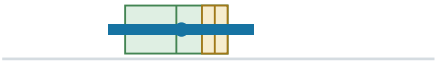
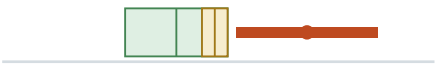
Synth · 1D-LTE Codex Grade · n=171		7.401 ±0.045
Synth · 1D-LTE · Gerber Codex Grade · n=171		7.403 ±0.045
EW · 1D-NLTE · Bergemann Codex Grade · n=5		7.433 ±0.123
Synth · 1D-NLTE · Bergemann Codex Grade · n=144		7.400 ±0.045
Synth · 1D-NLTE · Gerber Codex Grade · n=171		7.454 ±0.045
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=171		7.499 ±0.045
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=171		7.538 ±0.045
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.048

VIS — HARPS Reference Grade · full lab pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.22–7.66

Synth · 1D-LTE Reference Grade · n=167		7.346 ±0.045
Synth · 1D-LTE · Gerber Reference Grade · n=167		7.352 ±0.045
Synth · 1D-NLTE · Bergemann Reference Grade · n=157		7.346 ±0.045
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.398 ±0.045
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=167		7.446 ±0.045
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=167		7.479 ±0.045
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=100		7.500 ±0.045

VIS axis: 7.22 · 7.33 · 7.44 · 7.55 · 7.66 A(Fe) dex

red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Reference Grade · n=11		7.460 ±0.057
Synth · 1D-LTE · Gerber Reference Grade · n=11		7.464 ±0.057
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.472 ±0.056
Synth · 1D-NLTE · Gerber Reference Grade · n=11		7.507 ±0.054
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.519 ±0.058
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.562 ±0.055

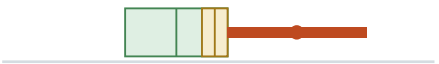
red-optical — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Codex Grade · n=97		7.437 ±0.049
Synth · 1D-LTE · Gerber Codex Grade · n=97		7.440 ±0.049
Synth · 1D-NLTE · Bergemann Codex Grade · n=46		7.449 ±0.048
Synth · 1D-NLTE · Gerber Codex Grade · n=97		7.487 ±0.049

red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Reference Grade · n=68		7.423 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=66		7.421 ±0.048
Synth · 1D-NLTE · Bergemann Reference Grade · n=34		7.418 ±0.048
Synth · 1D-NLTE · Gerber Reference Grade · n=68		7.455 ±0.048

red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Reference Grade · n=11		7.449 ±0.056
Synth · 1D-LTE · Gerber Reference Grade · n=11		7.451 ±0.056
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.488 ±0.058
Synth · 1D-NLTE · Gerber Reference Grade · n=11		7.503 ±0.053
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.525 ±0.057
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.554 ±0.054

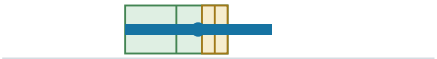
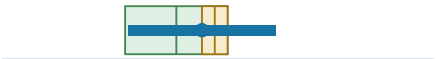
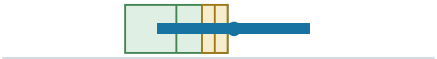
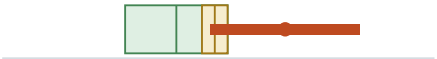
red-optical — Kitt Peak — 1984 Molecfit corrected Codex graded pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Codex Grade · n=66		7.413 ±0.048
Synth · 1D-LTE · Gerber Codex Grade · n=67		7.419 ±0.048
Synth · 1D-NLTE · Bergemann Codex Grade · n=33		7.403 ±0.049
Synth · 1D-NLTE · Gerber Codex Grade · n=65		7.454 ±0.048

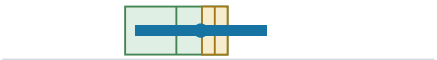
red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Reference Grade · n=66		7.413 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=67		7.419 ±0.048
Synth · 1D-NLTE · Bergemann Reference Grade · n=33		7.403 ±0.049
Synth · 1D-NLTE · Gerber Reference Grade · n=65		7.454 ±0.048

red-optical — IAG Reference Grade · Asplund replication solar_iag
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Reference Grade · n=11		7.477 ±0.057
Synth · 1D-LTE · Gerber Reference Grade · n=11		7.480 ±0.057
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.505 ±0.060
Synth · 1D-NLTE · Gerber Reference Grade · n=11		7.522 ±0.055
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.545 ±0.058
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.576 ±0.056

red-optical — IAG Codex graded pool solar_iag
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Codex Grade · n=96		7.437 ±0.051
Synth · 1D-LTE · Gerber Codex Grade · n=96		7.437 ±0.051
Synth · 1D-NLTE · Bergemann Codex Grade · n=46		7.446 ±0.056
Synth · 1D-NLTE · Gerber Codex Grade · n=95		7.479 ±0.051

red-optical — IAG Reference Grade · full lab pool solar_iag
A(Fe) dex; axis 7.32–7.66

Synth · 1D-LTE Reference Grade · n=67		7.424 ±0.051
Synth · 1D-LTE · Gerber Reference Grade · n=66		7.423 ±0.051
Synth · 1D-NLTE · Bergemann Reference Grade · n=33		7.422 ±0.062
Synth · 1D-NLTE · Gerber Reference Grade · n=66		7.454 ±0.051

red-optical axis: 7.32 · 7.41 · 7.49 · 7.58 · 7.66 A(Fe) dex

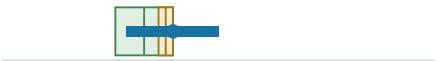
NIR — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Reference Grade · n=1		7.478 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.481 ±0.048
Synth · 1D-NLTE · Gerber Reference Grade · n=1		7.525 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.533 ±0.048
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.569 ±0.048

NIR — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Codex Grade · n=21		7.517 ±0.064
Synth · 1D-LTE · Gerber Codex Grade · n=21		7.508 ±0.064
Synth · 1D-NLTE · Bergemann Codex Grade · n=5		7.538 ±0.099

NIR — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Reference Grade · n=21		7.517 ±0.064
Synth · 1D-LTE · Gerber Reference Grade · n=21		7.500 ±0.064
Synth · 1D-NLTE · Bergemann Reference Grade · n=5		7.538 ±0.099
Synth · 1D-NLTE · Gerber Reference Grade · n=19		7.552 ±0.064
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.586 ±0.064
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=20		7.622 ±0.063

NIR — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Reference Grade · n=1		7.451 ±0.051
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.454 ±0.051
Synth · 1D-NLTE · Gerber Reference Grade · n=1		7.498 ±0.051
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.506 ±0.051
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.541 ±0.051

NIR — Kitt Peak — 1984 Molecfit corrected Codex graded pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Codex Grade · n=18		7.495 ±0.057
Synth · 1D-LTE · Gerber Codex Grade · n=20		7.482 ±0.056
Synth · 1D-NLTE · Bergemann Codex Grade · n=5		7.368 ±0.075

NIR — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Reference Grade · n=18		7.495 ±0.057
Synth · 1D-LTE · Gerber Reference Grade · n=20		7.478 ±0.068
Synth · 1D-NLTE · Bergemann Reference Grade · n=5		7.368 ±0.075
Synth · 1D-NLTE · Gerber Reference Grade · n=20		7.514 ±0.069
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=20		7.559 ±0.068
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=20		7.577 ±0.068

NIR — IAG Reference Grade · Asplund replication solar_iag
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Reference Grade · n=1		7.467 ±0.048
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.471 ±0.048
Synth · 1D-NLTE · Gerber Reference Grade · n=1		7.514 ±0.048
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.523 ±0.049
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.558 ±0.049

NIR — IAG Codex graded pool solar_iag
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Codex Grade · n=24		7.460 ±0.082
Synth · 1D-LTE · Gerber Codex Grade · n=23		7.468 ±0.085
Synth · 1D-NLTE · Bergemann Codex Grade · n=7		7.369 ±0.076

NIR — IAG Reference Grade · full lab pool solar_iag
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Reference Grade · n=17		7.520 ±0.105
Synth · 1D-LTE · Gerber Reference Grade · n=17		7.493 ±0.104
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.515 ±0.115
Synth · 1D-NLTE · Gerber Reference Grade · n=17		7.555 ±0.105
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=17		7.548 ±0.103
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=17		7.598 ±0.104

NIR — CRIRES+ Codex graded pool solar_crires_plus_y_wide_rya1054
A(Fe) dex; axis 7.26–7.86

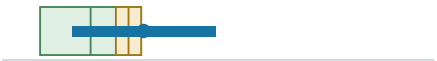
Synth · 1D-LTE Codex Grade · n=5		7.551 ±0.099
Synth · 1D-LTE · Gerber Codex Grade · n=5		7.552 ±0.098
Synth · 1D-NLTE · Bergemann Codex Grade · n=2		7.637 ±0.197

NIR — CRIRES+ Reference Grade · full lab pool solar_crires_plus_y_wide_rya1054
A(Fe) dex; axis 7.26–7.86

Synth · 1D-LTE Reference Grade · n=5		7.551 ±0.099
Synth · 1D-LTE · Gerber Reference Grade · n=5		7.557 ±0.091
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.637 ±0.197
Synth · 1D-NLTE · Gerber Reference Grade · n=5		7.594 ±0.102
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=5		7.607 ±0.087
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=5		7.635 ±0.099

NIR axis: 7.26 · 7.41 · 7.56 · 7.71 · 7.86 A(Fe) dex

H — CRIRES+ Codex graded pool solar_crires_plus_h_rya1232
A(Fe) dex; axis 7.39–7.73

Synth · 1D-LTE Codex Grade · n=19		7.497 ±0.057
Synth · 1D-LTE · Gerber Codex Grade · n=19		7.502 ±0.057
Synth · 1D-NLTE · Gerber Codex Grade · n=8		7.649 ±0.053

H — CRIRES+ Reference Grade · full lab pool solar_crires_plus_h_rya1232
A(Fe) dex; axis 7.39–7.73

Synth · 1D-LTE Reference Grade · n=19		7.497 ±0.057
Synth · 1D-LTE · Gerber Reference Grade · n=19		7.502 ±0.057

H — CRIRES+ Reference Grade · full lab pool solar_crires_plus_h_rya1232
A(Fe) dex; axis 7.39–7.73

Synth · 1D-NLTE · Gerber
Reference Grade · n=8



7.649
±0.053

H axis: 7.39 · 7.48 · 7.56 · 7.65 · 7.73 A(Fe) dex

How this was measured

Each row is one independent product. The result includes its published uncertainty (dex, Asplund 2021 convention).

Band	H
What was measured	Fe I · our-graded
Solar spectra	CRIRES+ solar_crires_plus_h_rya1232 corrected: ESO molecfit over EVERY H segment (116/116; H2O+CO2+CH4, per-night GDAS), rest-frame conditioned 7/7; pixels with transmission < 0.5 not written; known residual CH4 at 16656.0 A (~24%)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	19
Result A(Fe)	7.497 ± 0.057
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	H
What was measured	Fe I · our-graded
Solar spectra	CRIRES+ solar_crires_plus_h_rya1232 corrected: ESO molecfit over EVERY H segment (116/116; H2O+CO2+CH4, per-night GDAS), rest-frame conditioned 7/7; pixels with transmission < 0.5 not written; known residual CH4 at 16656.0 A (~24%)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Codex Grade
Lines	8
Result A(Fe)	7.649 ± 0.053
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	H
What was measured	Fe I · our-graded
Solar spectra	CRIRES+ solar_crires_plus_h_rya1232 corrected: ESO molecfit over EVERY H segment (116/116; H2O+CO2+CH4, per-night GDAS), rest-frame conditioned 7/7; pixels with transmission < 0.5 not written; known residual CH4 at 16656.0 A (~24%)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	19
Result A(Fe)	7.502 ± 0.057
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	H
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_h_rya1232 corrected: ESO molecfit over EVERY H segment (116/116; H2O+CO2+CH4, per-night GDAS), rest-frame conditioned 7/7; pixels with transmission < 0.5 not written; known residual CH4 at 16656.0 A (~24%)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	19
Result A(Fe)	7.497 ± 0.057
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	H
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_h_rya1232 corrected: ESO molecfit over EVERY H segment (116/116; H2O+CO2+CH4, per-night GDAS), rest-frame conditioned 7/7; pixels with transmission < 0.5 not written; known residual CH4 at 16656.0 A (~24%)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	8
Result A(Fe)	7.649 ± 0.053
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	H
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_h_rya1232 corrected: ESO molecfit over EVERY H segment (116/116; H2O+CO2+CH4, per-night GDAS), rest-frame conditioned 7/7; pixels with transmission < 0.5 not written; known residual CH4 at 16656.0 A (~24%)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	19
Result A(Fe)	7.502 ± 0.057
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	5
Result A(Fe)	7.551 ± 0.099
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit

Band	NIR
What was measured	Fe I · our-graded
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	2
Result A(Fe)	7.637 ± 0.197
Uncertainty and product notes	line-to-line scatter ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	5
Result A(Fe)	7.552 ± 0.098
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	5
Result A(Fe)	7.551 ± 0.099
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	2
Result A(Fe)	7.637 ± 0.197
Uncertainty and product notes	line-to-line scatter ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	5

Band	NIR
What was measured	Fe I · reference
Result A(Fe)	7.594 ± 0.102
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	5
Result A(Fe)	7.557 ± 0.091
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	5
Result A(Fe)	7.607 ± 0.087
Uncertainty and product notes	line-to-line scatter ξ MEASURED experimental, not adopted

Band	NIR
What was measured	Fe I · reference
Solar spectra	CRIRES+ solar_crires_plus_y_wide_rya1054 corrected: cr2res + molecfit
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	5
Result A(Fe)	7.635 ± 0.099
Uncertainty and product notes	line-to-line scatter ξ MEASURED experimental, not adopted

Band	NIR
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	1
Result A(Fe)	7.467 ± 0.048

Band	NIR
What was measured	Fe I · asplund
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	1
Result A(Fe)	7.514 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	1
Result A(Fe)	7.471 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	1
Result A(Fe)	7.523 ± 0.049
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	NIR
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	1

Band	NIR
What was measured	Fe I · asplund
Result A(Fe)	7.558 ± 0.049
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	24
Result A(Fe)	7.460 ± 0.082
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	7
Result A(Fe)	7.369 ± 0.076
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	23
Result A(Fe)	7.468 ± 0.085
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	17
Result A(Fe)	7.520 ± 0.105
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	4
Result A(Fe)	7.515 ± 0.115
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	17
Result A(Fe)	7.555 ± 0.105
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	17
Result A(Fe)	7.493 ± 0.104
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	17
Result A(Fe)	7.548 ± 0.103
Uncertainty and product notes	line-to-line scatter ξ MEASURED experimental, not adopted

Band	NIR
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade

Band	NIR
What was measured	Fe I · reference
Lines	17
Result A(Fe)	7.598 ± 0.104
Uncertainty and product notes	line-to-line scatter ξ MEASURED experimental, not adopted

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	1
Result A(Fe)	7.478 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	1
Result A(Fe)	7.525 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	1
Result A(Fe)	7.481 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade

Band	NIR
What was measured	Fe I · asplund
Lines	1
Result A(Fe)	7.533 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	1
Result A(Fe)	7.569 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	21
Result A(Fe)	7.517 ± 0.064
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	5
Result A(Fe)	7.538 ± 0.099
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade

Band	NIR
What was measured	Fe I · our-graded
Lines	21
Result A(Fe)	7.508 ± 0.064
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	21
Result A(Fe)	7.517 ± 0.064
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	5
Result A(Fe)	7.538 ± 0.099
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	19
Result A(Fe)	7.552 ± 0.064
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	21
Result A(Fe)	7.500 ± 0.064
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	22
Result A(Fe)	7.586 ± 0.064
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED experimental, not adopted

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	20
Result A(Fe)	7.622 ± 0.063
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED experimental, not adopted

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	1
Result A(Fe)	7.451 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	1
Result A(Fe)	7.498 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	1
Result A(Fe)	7.454 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	1
Result A(Fe)	7.506 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	NIR
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	1
Result A(Fe)	7.541 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	18
Result A(Fe)	7.495 ± 0.057
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	5
Result A(Fe)	7.368 ± 0.075
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	20
Result A(Fe)	7.482 ± 0.056
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	18
Result A(Fe)	7.495 ± 0.057
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	5
Result A(Fe)	7.368 ± 0.075
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	20

Band	NIR
What was measured	Fe I · reference
Result A(Fe)	7.514 ± 0.069
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	20
Result A(Fe)	7.478 ± 0.068
Uncertainty and product notes	line-to-line scatter ξ MEASURED

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	20
Result A(Fe)	7.559 ± 0.068
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED experimental, not adopted

Band	NIR
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	20
Result A(Fe)	7.577 ± 0.068
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe I · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Reference Grade
Lines	21
Result A(Fe)	7.477 ± 0.049
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	27
Result A(Fe)	7.398 ± 0.047
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe I · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	20
Result A(Fe)	7.417 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe I · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	27
Result A(Fe)	7.452 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe I · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	27
Result A(Fe)	7.400 ± 0.047
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe I · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	27
Result A(Fe)	7.480 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · asplund
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	27
Result A(Fe)	7.515 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	109
Result A(Fe)	7.339 ± 0.087
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	105
Result A(Fe)	7.350 ± 0.088
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	106
Result A(Fe)	7.419 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	102
Result A(Fe)	7.429 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	EW · 1D-LTE · PROFILEFIT · 1D-LTE · Codex Grade
Lines	6
Result A(Fe)	7.498 ± 0.120
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	EW · 1D-NLTE · Bergemann · PROFILEFIT · ENGINE-A · Codex Grade
Lines	5
Result A(Fe)	7.433 ± 0.123
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	EW · 1D-LTE · PROFILEFIT · ENGINE-B · Codex Grade
Lines	5

Band	VIS
What was measured	Fe I · our-graded
Result A(Fe)	7.431 ± 0.122
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	171
Result A(Fe)	7.401 ± 0.045
Uncertainty and product notes	gf scale (UNGRADED) ξ ALIASED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Codex Grade
Lines	50
Result A(Fe)	7.512 ± 0.048
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	144
Result A(Fe)	7.400 ± 0.045
Uncertainty and product notes	gf scale (UNGRADED) ξ ALIASED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Codex Grade
Lines	171
Result A(Fe)	7.454 ± 0.045
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	171
Result A(Fe)	7.403 ± 0.045
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Codex Grade
Lines	171
Result A(Fe)	7.499 ± 0.045
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Codex Grade
Lines	171
Result A(Fe)	7.538 ± 0.045
Uncertainty and product notes	gf scale (UNGRADED) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	167
Result A(Fe)	7.346 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Reference Grade
Lines	100
Result A(Fe)	7.500 ± 0.045
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	157
Result A(Fe)	7.346 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	167
Result A(Fe)	7.398 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	167
Result A(Fe)	7.352 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	167

Band	VIS
What was measured	Fe I · reference
Result A(Fe)	7.446 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · reference
Solar spectra	HARPS solar_harps_molecfit_corrected corrected: ESO molecfit 4.4.4 per-exposure, observation-night GDAS (RYA-931)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	167
Result A(Fe)	7.479 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ NOT_IN_CAMPAIGN INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Reference Grade
Lines	21
Result A(Fe)	7.471 ± 0.046
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	22
Result A(Fe)	7.443 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	18

Band	VIS
What was measured	Fe I · asplund
Result A(Fe)	7.462 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	22
Result A(Fe)	7.484 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	22
Result A(Fe)	7.450 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	22
Result A(Fe)	7.531 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	22
Result A(Fe)	7.546 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	106
Result A(Fe)	7.329 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	101
Result A(Fe)	7.340 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	EW · 1D-LTE · PROFILEFIT · 1D-LTE · Codex Grade
Lines	13
Result A(Fe)	7.445 ± 0.104
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	EW · 1D-NLTE · Bergemann · PROFILEFIT · ENGINE-A · Codex Grade
Lines	12
Result A(Fe)	7.455 ± 0.107
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	EW · 1D-LTE · PROFILEFIT · ENGINE-B · Codex Grade
Lines	9

Band	VIS
What was measured	Fe I · our-graded
Result A(Fe)	7.479 ± 0.090
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	65
Result A(Fe)	7.410 ± 0.050
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Codex Grade
Lines	50
Result A(Fe)	7.505 ± 0.047
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	59
Result A(Fe)	7.411 ± 0.050
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Codex Grade
Lines	67
Result A(Fe)	7.488 ± 0.066

Band	VIS
What was measured	Fe I · our-graded
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	67
Result A(Fe)	7.439 ± 0.067
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Codex Grade
Lines	67
Result A(Fe)	7.541 ± 0.067
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Codex Grade
Lines	67
Result A(Fe)	7.562 ± 0.066
Uncertainty and product notes	gf scale (cited lab) ξ UNMEASURED INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. experimental, not adopted

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade

Band	VIS
What was measured	Fe I · reference
Lines	166
Result A(Fe)	7.367 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Reference Grade
Lines	101
Result A(Fe)	7.485 ± 0.045
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	156
Result A(Fe)	7.371 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	167
Result A(Fe)	7.430 ± 0.044
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	167
Result A(Fe)	7.373 ± 0.044
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Reference Grade
Lines	21
Result A(Fe)	7.504 ± 0.046
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	22
Result A(Fe)	7.455 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	18
Result A(Fe)	7.469 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	22
Result A(Fe)	7.495 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	22

Band	VIS
What was measured	Fe I · asplund
Result A(Fe)	7.462 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	22
Result A(Fe)	7.542 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	22
Result A(Fe)	7.562 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	106
Result A(Fe)	7.347 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Deep Grade
Lines	102
Result A(Fe)	7.357 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	EW · 1D-LTE · PROFILEFIT · 1D-LTE · Codex Grade
Lines	13
Result A(Fe)	7.463 ± 0.111
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	EW · 1D-NLTE · Bergemann · PROFILEFIT · ENGINE-A · Codex Grade
Lines	12
Result A(Fe)	7.464 ± 0.112
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	65
Result A(Fe)	7.414 ± 0.049
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Codex Grade
Lines	50
Result A(Fe)	7.512 ± 0.047
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	59

Band	VIS
What was measured	Fe I · our-graded
Result A(Fe)	7.420 ± 0.049
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Codex Grade
Lines	65
Result A(Fe)	7.467 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	65
Result A(Fe)	7.415 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Codex Grade
Lines	65
Result A(Fe)	7.511 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Codex Grade
Lines	65
Result A(Fe)	7.548 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED experimental, not adopted

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	167
Result A(Fe)	7.368 ± 0.044
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 3D-NLTE · Amarsi · lab-gf · SYNTH · ENGINE-A-3DNLTE · Reference Grade
Lines	101
Result A(Fe)	7.492 ± 0.045
Uncertainty and product notes	budget not attached ξ NOT_APPLICABLE

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	157
Result A(Fe)	7.377 ± 0.045
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	167
Result A(Fe)	7.432 ± 0.044
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	VIS
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	166

Band	VIS
What was measured	Fe I · reference
Result A(Fe)	7.378 ± 0.044
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	near-UV
What was measured	Fe I · our-deep-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Deep Grade
Lines	55
Result A(Fe)	7.554 ± 0.075
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	48
Result A(Fe)	7.515 ± 0.072
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	48
Result A(Fe)	7.523 ± 0.072
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	51
Result A(Fe)	7.559 ± 0.067
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	51
Result A(Fe)	7.494 ± 0.067
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	52
Result A(Fe)	7.588 ± 0.070
Uncertainty and product notes	pseudo-continuum ξ MEASURED experimental, not adopted

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	53
Result A(Fe)	7.678 ± 0.069
Uncertainty and product notes	pseudo-continuum ξ MEASURED experimental, not adopted

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	46
Result A(Fe)	7.380 ± 0.075
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade

Band	near-UV
What was measured	Fe I · reference
Lines	46
Result A(Fe)	7.389 ± 0.075
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	51
Result A(Fe)	7.452 ± 0.065
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	50
Result A(Fe)	7.398 ± 0.065
Uncertainty and product notes	pseudo-continuum ξ MEASURED

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	51
Result A(Fe)	7.500 ± 0.067
Uncertainty and product notes	pseudo-continuum ξ MEASURED experimental, not adopted

Band	near-UV
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	51
Result A(Fe)	7.543 ± 0.062

Band	near-UV
What was measured	Fe I · reference
Uncertainty and product notes	pseudo-continuum ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	11
Result A(Fe)	7.477 ± 0.057
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	4
Result A(Fe)	7.505 ± 0.060
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	11
Result A(Fe)	7.522 ± 0.055
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	11
Result A(Fe)	7.480 ± 0.057
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	11
Result A(Fe)	7.545 ± 0.058
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	11
Result A(Fe)	7.576 ± 0.056
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	96
Result A(Fe)	7.437 ± 0.051
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	46
Result A(Fe)	7.446 ± 0.056
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Codex Grade

Band	red-optical
What was measured	Fe I · our-graded
Lines	95
Result A(Fe)	7.479 ± 0.051
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	96
Result A(Fe)	7.437 ± 0.051
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	67
Result A(Fe)	7.424 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	33
Result A(Fe)	7.422 ± 0.062
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	66
Result A(Fe)	7.454 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	IAG solar_iag corrected: as delivered — measured telluric-free to 0.1% in the O2 A-band core (RYA-783/786)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	66
Result A(Fe)	7.423 ± 0.051
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	11
Result A(Fe)	7.460 ± 0.057
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	4
Result A(Fe)	7.472 ± 0.056
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	11
Result A(Fe)	7.507 ± 0.054
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	11

Band	red-optical
What was measured	Fe I · asplund
Result A(Fe)	7.464 ± 0.057
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	11
Result A(Fe)	7.519 ± 0.058
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	11
Result A(Fe)	7.562 ± 0.055
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	97
Result A(Fe)	7.437 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade
Lines	46
Result A(Fe)	7.449 ± 0.048
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Codex Grade
Lines	97
Result A(Fe)	7.487 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	97
Result A(Fe)	7.440 ± 0.049
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	68
Result A(Fe)	7.423 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	34
Result A(Fe)	7.418 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	68

Band	red-optical
What was measured	Fe I · reference
Result A(Fe)	7.455 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_kurucz2005_corrected corrected: Kurucz-2005 product is delivered telluric-corrected
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	66
Result A(Fe)	7.421 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	11
Result A(Fe)	7.449 ± 0.056
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	4
Result A(Fe)	7.488 ± 0.058
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	11
Result A(Fe)	7.503 ± 0.053
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	11
Result A(Fe)	7.451 ± 0.056
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-LTE · Gerber · stagger · SYNTH · synth-mean3D-LTE-gerber-stagger · Reference Grade
Lines	11
Result A(Fe)	7.525 ± 0.057
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe I · asplund
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · <3D>-NLTE · Gerber · stagger · SYNTH · synth-mean3D-NLTE-gerber-stagger · Reference Grade
Lines	11
Result A(Fe)	7.554 ± 0.054
Uncertainty and product notes	gf scale (UNGRADED) ξ MEASURED experimental, not adopted

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Codex Grade
Lines	66
Result A(Fe)	7.413 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Codex Grade

Band	red-optical
What was measured	Fe I · our-graded
Lines	33
Result A(Fe)	7.403 ± 0.049
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Codex Grade
Lines	65
Result A(Fe)	7.454 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · our-graded
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Codex Grade
Lines	67
Result A(Fe)	7.419 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · SYNTH · 1D-LTE · Reference Grade
Lines	66
Result A(Fe)	7.413 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · Reference Grade
Lines	33
Result A(Fe)	7.403 ± 0.049
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-NLTE · Gerber · SYNTH · ENGINE-B-NLTE · Reference Grade
Lines	65
Result A(Fe)	7.454 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Band	red-optical
What was measured	Fe I · reference
Solar spectra	Kitt Peak solar_kpno_molecfit_corrected corrected: ESO molecfit 4.4.4, six registered bands (RYA-940)
Treatments	Synth · 1D-LTE · Gerber · SYNTH · synth-1D-LTE-gerber · Reference Grade
Lines	67
Result A(Fe)	7.419 ± 0.048
Uncertainty and product notes	gf scale (cited lab) ξ MEASURED

Solar iron context

Fe I supplies the bulk of the neutral-iron line statistics used for the Solar abundance reference. Separate instruments, line pools and treatments test the stability of that reference; their differences remain visible.

Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline. The headline is the per-band best-available product, selected from this feed by the publication generator rather than fixed as a constant: RYA-850 promoted the graded laboratory-gf pool to the PRIMARY reported value, and RYA-819 retired the earlier constructed anchor after showing its Magic-2013 correction wrong in both magnitude and shape. RYA-161: nothing calibrated to a target.

Frankenstein — experimental, pending the Bride

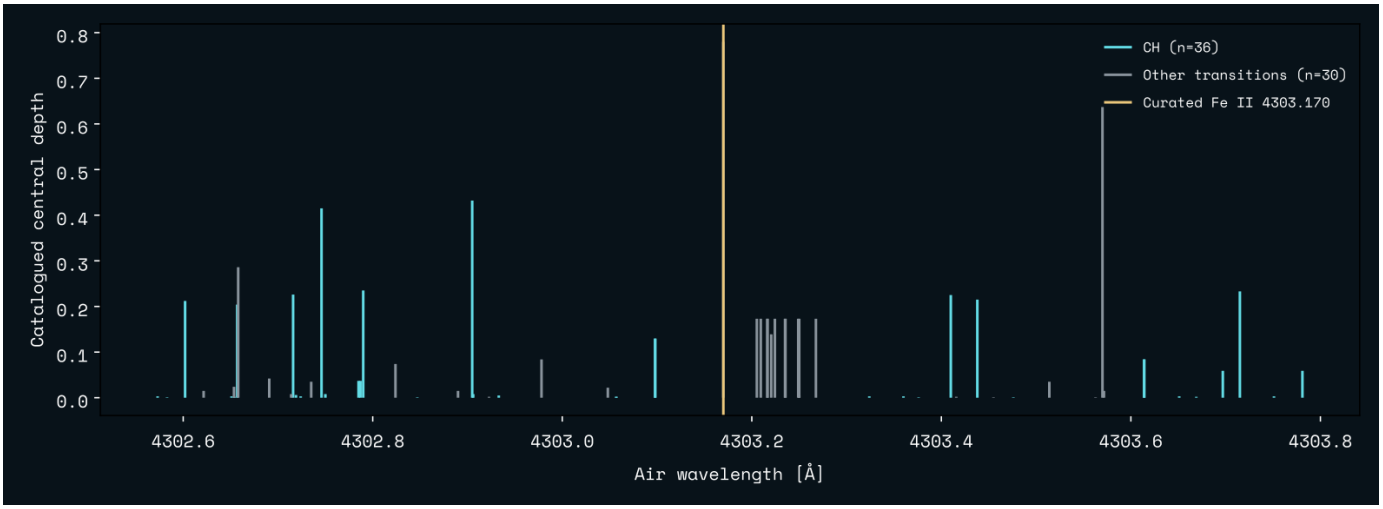
48 experimental products remain available in the Fe I product values and downloads.

RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR

is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

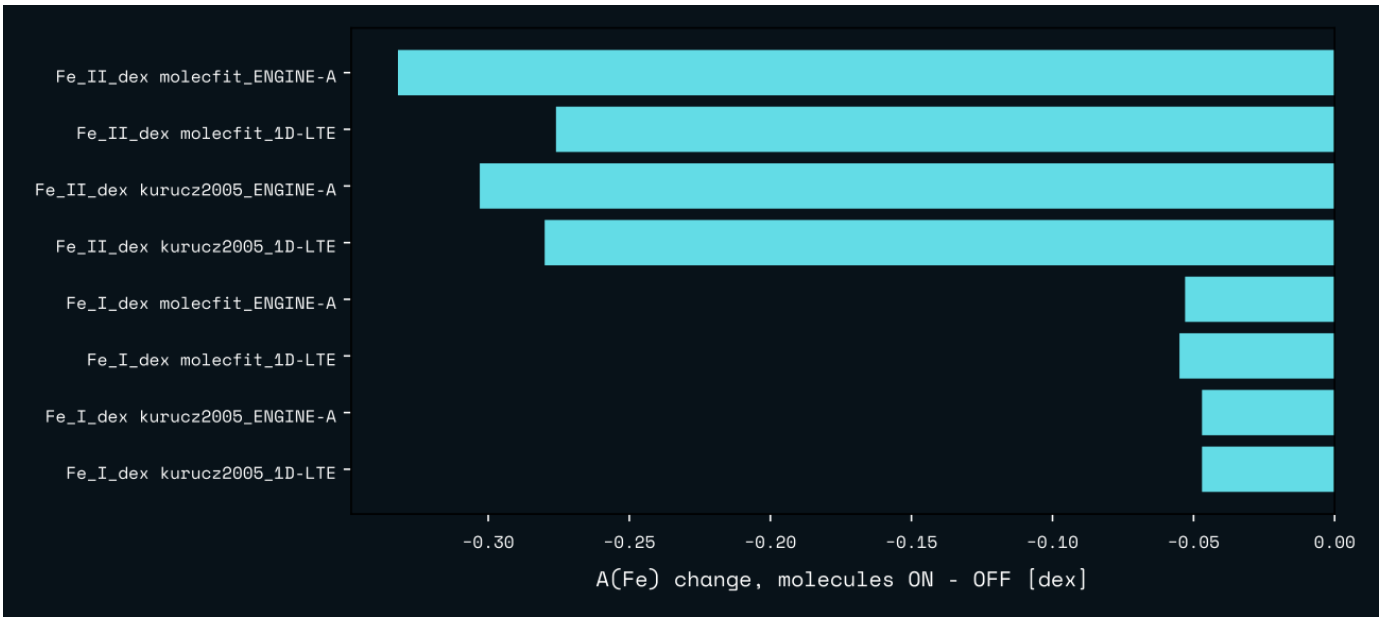
Problem-line and model diagnostics

CH G-band artifact



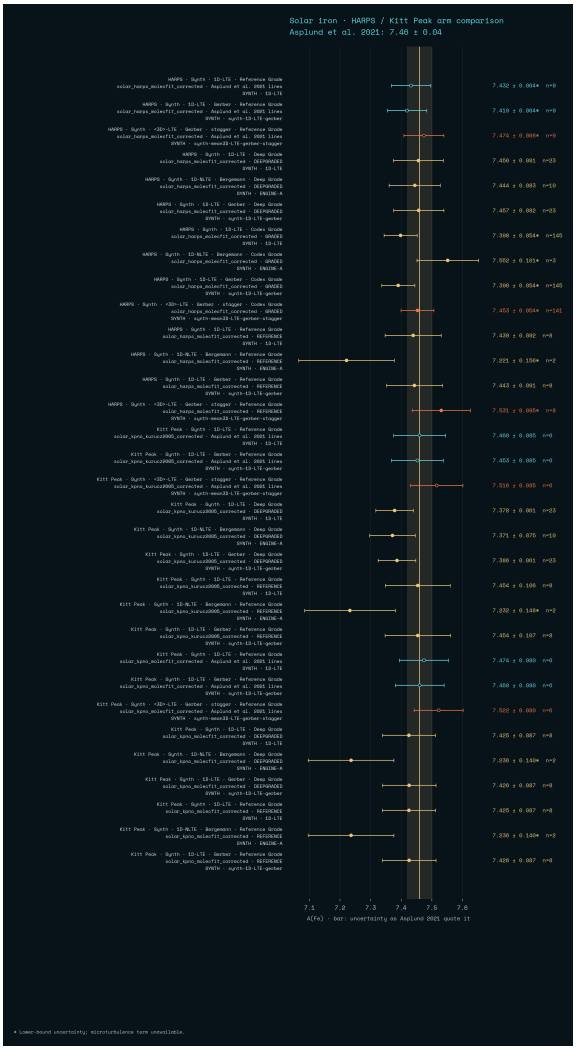
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · A = 10.0 (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 A fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. Source evidence

Near-UV molecular opacity



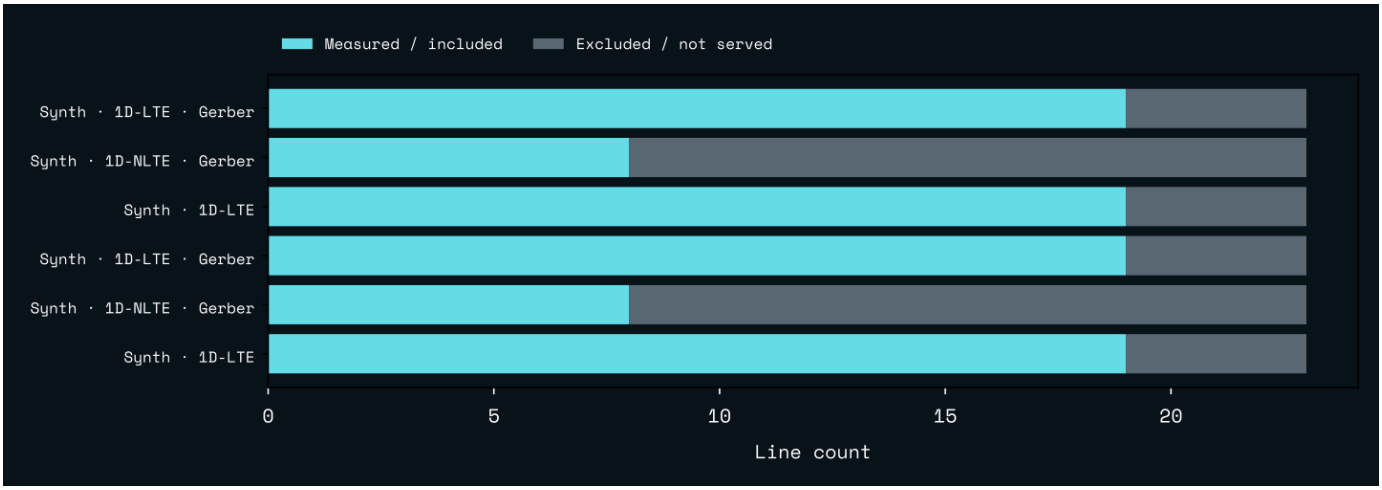
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. Source evidence

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. Source evidence

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. Source evidence

Download the evidence

1341 graded per-line measurements exported. Evidence for 152 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

Graded per-line abundances (CSV)

All product fields (CSV; nested fields preserved as JSON)

Per-product line-evidence coverage (CSV)

Source Fe.json

Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0
Science source commit 28f3e442ab200d01696a8074dd7c4e1186a3dedb
Fe.json v1.575 · feed timestamp 2026-11-06T22:35:23Z
Generated 2026-11-06T22:28:38+00:00
Feed SHA-256 33703076fda06b417df0cb77e707254006228e48c54c0439dbb4cf89fe4c78d1

Refreshed product tracker · Build manifest · Fe II appendix

References

Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision (doi: 10.1051/0004-6361/202140445). AGSS21 Solar reference and the green literature comparison; Table A.2 defines the Reference Grade Fe I line set.

Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Amarsi, A. M.; Liljegren, S.; Nissen, P. E. (2022). 3D non-LTE iron abundances in FG-type dwarfs (doi: 10.1051/0004-6361/202244542). The Amarsi 3D non-LTE iron correction grid used by the displayed products.

Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum (doi: 10.1051/0004-6361/202243673). The Gerber Turbospectrum non-LTE synthesis implementation named in the model rows.

Den Hartog, E. A.; Ruffoni, M. P.; Lawler, J. E.; Pickering, J. C.; Lind, K.; Brewer, N. R. (2014). Fe I Oscillator Strengths for Transitions from High-lying Even-parity Levels (doi: 10.1088/0067-0049/215/2/23). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Brewer, N. R.; Lind, K.; Nave, G.; Pickering, J. C. (2014). Fe I oscillator strengths for the Gaia-ESO survey (doi: 10.1093/mnras/stu780). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Belmonte, M. T.; Pickering, J. C.; Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Guzman, A.; Heiter, U. (2017). Fe I Oscillator Strengths for Transitions from High-lying Odd-parity Levels (doi: 10.3847/1538-4357/aa8cd3). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.